

⚡ PRODUCTION-GRADE GOVTECH PLATFORM

CivicPulse — Autonomous Municipal Grievance Redressal

Transforming legacy urban grievance management into a sub-second, closed-loop AI triage and automated spatial dispatch grid.

EXECUTIVE PRESENTERS

 Avinash Peela

 Praveen

 Navadeep

 Ritul

OPERATIONAL SCOPE

Zero triage delay bridging Citizens, Ward Officers, Field Crews & Swachh Survekshan Quality Auditors.

⚠️ LEGACY MUNICIPAL PORTALS

- ✗ **Multi-Day Triage Delays:** Manual paper routing and departmental silos cause massive backlog accumulation.
- ✗ **Static Ticketing Queues:** Absence of real-time GPS tracking leaves field repair contractors unmonitored.
- ✗ **Broken Feedback Loops:** Citizens receive zero proof-of-work, leading to duplicate complaints and mistrust.

🛡️ THE CIVICPULSE GRID

- ✓ **Instant AI Hazard Classification:** Multimodal Gemini Vision categorizes issues in <1s on upload.
- ✓ **Live GIS Spatial Dispatch:** Dynamic fleet routing and active SLA countdowns enforced at ward levels.
- ✓ **Closed-Loop Photo Verification:** Geo-tagged resolution photos and citizen quality scorecards ensure accountability.

STEP 01

1-Click Auth

Google SSO or mobile OTP login lands citizen directly on a personalized ward dashboard with zero onboarding friction.

STEP 02

Photo Capture

1-tap camera trigger to snap raw civic hazards: potholes, open drains, water bursts, or illegal garbage piles.

STEP 03

GPS Snapping

Device grabs high-precision coordinates, snapping the issue directly to the corresponding municipal Ward and street address.

STEP 04

Canvas Compression

Client-side HTML5 engine compresses payload to <100 KB, guaranteeing sub-second uploads even on poor 3G/4G networks.

STEP 05

Live Tracking

Citizen receives a live ticket ID with active SLA timer, crew assignment status, and stage milestones (Assigned → Fixed).

VISION PIPELINE



Multimodal Hazard Identification

Images ingested through Gemini Vision are dynamically classified by structural hazard type (Public Works, Sanitation, Electrical, Water Supply).

🎯 **Zero-Triage Routing:** Matches hazard class directly to the designated municipal department fleet with zero manual clerk intervention.

TRIAGE MATRIX



Automated Severity Scoring

P1 CRITICAL

Immediate physical threat (deep potholes, open drains, cave-ins).
24h SLA

P2 URGENT

High municipal disruption (water main burst, broken traffic signal).
48h SLA

P3 NORMAL

Routine maintenance (minor cracks, faded signs, street lamps). **7d**
SLA

SUB-SECOND SYNC



Desk WebSocket Ingestion

Direct Firestore document listeners (onSnapshot) eliminate heavy HTTP polling. Citizen filings instantly manifest on Officer Tactical GIS Desks across the city.

SPATIAL ROUTING



Field Fleet Dispatch

Ward officers dispatch the nearest available repair fleet unit with dynamic Google Maps trajectory polylines, density clustering, and turn-by-turn navigation.

AUDIT PROTOCOL



Proof-of-Fix Verification

Field contractors must upload timestamped, GPS-verified resolution photos directly on-site before tickets can transition to resolved status.

LEVEL 1

Field Contractor / JE

Initial Repair Window

Receives direct push dispatch and trajectory routing. Responsible for physical execution within standard resolution window (P1: 24h, P2: 48h, P3: 7d).

LEVEL 2

Ward Sanitary Inspector / AE

75% SLA Auto-Alert

Automatically alerted if a ticket nears 75% of SLA threshold without crew assignment. System promotes priority (P3 → P2 → P1) with high-visibility dashboard flags.

LEVEL 3

Municipal Commissioner / Apex HQ

Apex Escalation & Penalty

Breached tickets bypass ward controls into Apex Command for inquiry. Penalty points directly impact the ward's official Swachh Survekshan audit scorecard.

SYSTEM LAYER	SELECTED STACK	KEY TECHNICAL ADVANTAGE
EDGE CLIENT	React 18 • TS • Tailwind • Vite	Sub-300ms TTI, strictly typed data contracts, and mobile-first responsiveness.
EDGE HOSTING	Cloudflare Pages & Global CDN	Zero server cost, infinite instant concurrency, and sub-50ms asset delivery.
REAL-TIME DB	Firestore (civictracker)	Sub-second sync via persistent WebSocket listeners with ACID transactional writes.
SECURITY & AUTH	Firebase Auth & Granular RBAC	Role separation for Citizens, Field Crews, Inspectors, Ward Officers, and Admins.
SPATIAL & GIS	Google Maps API & Spatial Layers	GPS coordinate snapping, SBM public facility routing, and ward heatmaps.

SERVERLESS



Zero Compute Overhead

Eliminates expensive fixed virtual machines and container orchestration by adopting a modern JAMstack serverless edge distribution architecture.

URBAN SCALE



Surge Concurrency

Effortlessly handles massive citizen traffic spikes during municipal cleanup drives, seasonal monsoons, or civic emergencies without throttling or cold starts.

ULTRA-LOW LATENCY



Sub-50ms Global PoPs

Static assets, client bundles, and WASM binaries cache directly at edge nodes across hundreds of global PoPs for instantaneous mobile loading.

GIS LOCATOR



Sanitation Locator

Interactive full-page directory mapping all public toilets, waste segregation centers, and SBM sanitation facilities across municipal wards.

DYNAMIC PROXIMITY



Real-Time Distance

Calculates walking and driving distance dynamically from the citizen's live GPS coordinates with integrated navigation routing.

CLEANLINESS AUDIT



Citizen Scorecards

Crowdsourced cleanliness ratings and hygiene feedback flow directly into municipal ward evaluation dashboards for contractor accountability.

MEASURED IMPACT



Sub-Second Triage

Reduces municipal grievance triage time from 3-5 business days to instant automated AI classification, enforcing ward-level transparency.

FISCAL EFFICIENCY



Near-Zero Run Cost

Architected on high-performance serverless tiers, keeping public software maintenance and cloud hosting bills near zero for smart cities.

NEXT-GEN IOT



Sensors & Vernacular

Planned integrations for smart drain fill-level IoT telemetry, multi-lingual vernacular voice grievance capture, and inter-city cluster analytics.

 OPEN FOR DISCUSSION

CivicPulse — Municipal Intelligence Grid

Empowering citizens, optimizing field operations, and enforcing urban governance through real-time autonomous technology.

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